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CerebroNet: A systematically derived explainable brain tumor classifier for resource-constrained MRI diagnostics

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ABSTRACT

While Artificial Intelligence (AI) has demonstrated potential in assisting MRI-based brain tumor diagnosis, its real-world utility appears to be limited by high computational demands and demographic biases inherent in Western-centric datasets. This study proposes a systematic optimization framework to develop a resource-efficient, explainable classifier tailored to an underrepresented South Asian cohort. We initially benchmarked 21 deep learning architectures to isolate the experimentally optimal teacher model. Subsequently, “CerebroNet” was derived via architectural ablation and refined through a holistic pipeline integrating knowledge distillation and unstructured pruning. The resulting model contains merely 0.637 million parameters, representing a 17-fold reduction relative to state-of-the-art benchmarks. Despite this substantial compression, CerebroNet seems to retain over 96% of the teacher’s diagnostic fidelity, attaining 95.79% accuracy on augmented data and 96.21% on raw clinical scans. Validation on an external dataset yielded 91% accuracy, suggesting a degree of robust generalization. Furthermore, explainable AI (Layer-CAM) analysis indicated that the student model likely preserves the relevant visual reasoning of the teacher. By balancing computational efficiency with demographic inclusivity, this work attempts to offer a reproducible strategy for implementing reliable AI diagnostics in resource-constrained healthcare environments.

Introduction

Brain tumors present a significant health challenge worldwide. A patient’s outcome and the success of treatment often rely on early detection and the precise identification of the tumor’s type, grade, and location¹. However, achieving a precise diagnosis remains arduous due to the diverse appearance of tumors and how their symptoms often overlap with those of other neurological conditions. Although Magnetic Resonance Imaging (MRI) is the main non-invasive diagnostic tool, its interpretation is subjective and seems to vary between physicians². These challenges underscore what seems to be a critical necessity for automated clinical support tools capable of enhancing decision-making consistency.

The domain has rapidly evolved from traditional machine learning approaches^{3–5} to advanced deep learning frameworks. Convolutional Neural Networks (CNNs)⁶ and Vision Transformers (ViTs)^{7,8} have shown diagnostic abilities that are said to be similar to those of human experts⁹. Despite these advancements, clinical adoption in resource-constrained settings appears restricted by hardware limitations, as state-of-the-art models often demand computational infrastructure unavailable in developing regions¹⁰. Furthermore, existing models largely rely on Western datasets, introducing algorithmic bias that may compromise reliability when applied to underrepresented ethnic groups, including South Asians^{11,12}. Addressing these disparities is essential, as ignoring them could limit the global generalization and trustworthiness of diagnostic AI¹³.

To close the gap between high-performance needs and limited resources while tackling demographic differences, this study offers a complete optimization framework. We propose a knowledge-centric approach that distills diagnostic capability from a computationally complex teacher into a streamlined student model through rigorous benchmarking, ablation study, and pruning. Crucially, we validate this framework using the “PMRAM: Bangladeshi Brain Cancer - MRI Dataset”¹⁴ to explicitly target the representation gap for South Asian populations. Given that South Asian genomic research indicates considerable ancestral diversity^{15,16}, validating on a Bangladeshi cohort exposes the model to a heterogeneous demographic profile, potentially strengthening the argument for broader applicability. Fig. 1 gives an overview of our study. The contributions of this work are as follows:

- **Evidence-Based Teacher Selection:** A robust performance baseline was established by benchmarking 21 advanced deep learning architectures on the Bangladeshi dataset, identifying EfficientNetB2 as the likely optimal teacher for knowledge transfer.

- **Systematic Architectural Derivation:** Rather than arbitrarily designing a lightweight model, a systematic ablation study was employed to derive CerebroNet, a minimal CNN architecture (0.637M parameters). This process isolated the most effective efficiency mechanisms, specifically channel attention, while eliminating what appeared to be redundant spatial modules.
- **Holistic Optimization Pipeline:** A rigorous optimization pipeline combining Knowledge Distillation (KD)¹⁷ with unstructured pruning¹⁸ was implemented. This pipeline successfully transferred over 96% of the teacher's diagnostic capability to the student model, achieving a final test accuracy of 95.79%.
- **Robust Cross-Population Validation and Interpretability:** To ensure clinical trust and generalization, the model was evaluated on both raw, un-augmented local data and an external public dataset. Furthermore, Layer-CAM¹⁹ (Explainable AI) was utilized to verify that the student model retains the salient visual reasoning of its teacher.

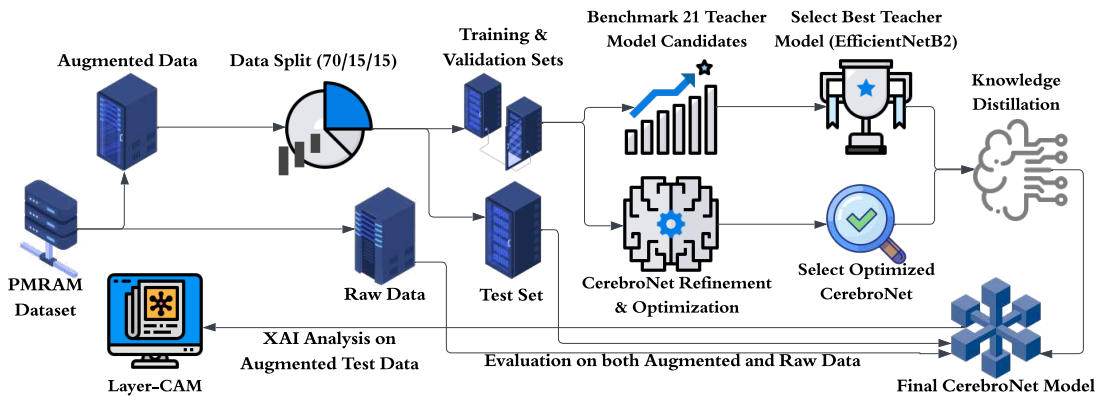


Figure 1. Overview of the study. The process begins with benchmarking 21 teacher models on the PMRAM dataset to select the best teacher. Concurrently, CerebroNet is designed and optimized using an ablation study. Finally, the best teacher transfers its knowledge to the optimized CerebroNet via knowledge distillation, resulting in the final, proposed model with XAI analysis.

Related work

Deep learning architectures and resource constraints

Initial implementations leveraged Residual Networks (ResNets) and ensemble strategies to improve diagnostic precision^{9,20}. Subsequent investigations prioritized deep, high-parameter architectures to minimize error rates⁶. For example, He et al.²¹ employed Grid Search Enhanced with Coordinate Ascent (GSECA) optimization on a ResNet-18 backbone using the Bangladeshi PMRAM dataset, attaining 95.69% accuracy. However, reliance on architectures exceeding 11 million parameters constrains deployment in resource-limited environments, emphasizing the need for hardware-optimized designs that sustain diagnostic accuracy without extensive computational demands.

Attention mechanisms and transformers

Following breakthroughs in natural language processing²², medical imaging research has increasingly incorporated Vision Transformers (ViTs) and advanced attention modules^{7,8}. Spatial and channel-wise attention mechanisms are frequently applied to accentuate relevant feature regions while suppressing background noise^{23–25}. Despite these developments, the assumption that complex attention integration inherently enhances lightweight model performance remains largely empirically unverified in constrained settings. A rigorous evaluation of specific attention types is necessary to determine if simpler mechanisms can deliver comparable results with reduced computational load.

Knowledge distillation and explainability

Knowledge Distillation (KD) has been utilized to compress models for medical applications, including tumor segmentation and classification^{17,26–28}. While Anantathanavit et al.¹⁰ illustrated the utility of KD for developing lightweight tools, its integration within a systematic optimization framework involving architectural derivation and pruning remains underutilized. Additionally,

the “black box” nature of deep learning mandates interpretability. Explainable AI (XAI) techniques like Class Activation Mapping (CAM) and Layer-CAM are being used more often to show how models make decisions^{29–33}. Although recent studies have applied XAI for architectural simplification^{34,35}, the rigorous verification of visual reasoning transfer, confirming the student model replicates the teacher’s pathological focus, is often neglected.

Materials and methods

Data

This study employs the “PMRAM: Bangladeshi Brain Cancer - MRI Dataset”¹⁴, a publicly accessible repository containing MRI scans collected from various medical institutions in Bangladesh. It consists of 1,505 raw MRI scans, comprising 373 Glioma, 363 Meningioma, 396 No Tumor, and 373 Pituitary tumor samples. The creators of the PMRAM dataset performed both augmentation, resulting in a balanced collection of 6,004 images (1,501 per class), and standardization of all images to a resolution of 512×512 for improved compatibility. Collected from diverse hospital sources across Bangladesh, this dataset introduces essential demographic diversity into the study. This dataset is particularly significant given the scarcity of high-quality medical imaging data in the context of brain cancer in Bangladesh. Representative samples are shown in Fig. 2.

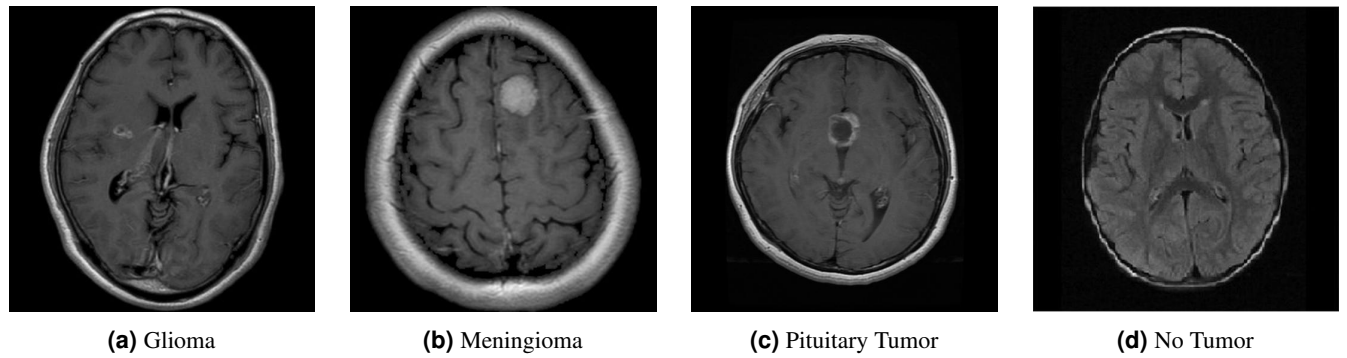


Figure 2. Representative MRI scans from the four diagnostic classes in the PMRAM dataset. The diversity in tumor location and appearance presents a significant challenge for automated classification.

We, for this study, resized the images to 256×256 pixels and then center-cropped them to 224×224 pixels. Normalization was executed using standard ImageNet mean and standard deviation values. We split the augmented dataset into training (70%), validation (15%), and testing (15%) subsets. Also, t-Distributed Stochastic Neighbor Embedding³⁶ (t-SNE) was conducted (Fig. 3) to inspect the data distribution. It appears that the augmented dataset manifests a denser and more continuous manifold structure relative to the raw data.

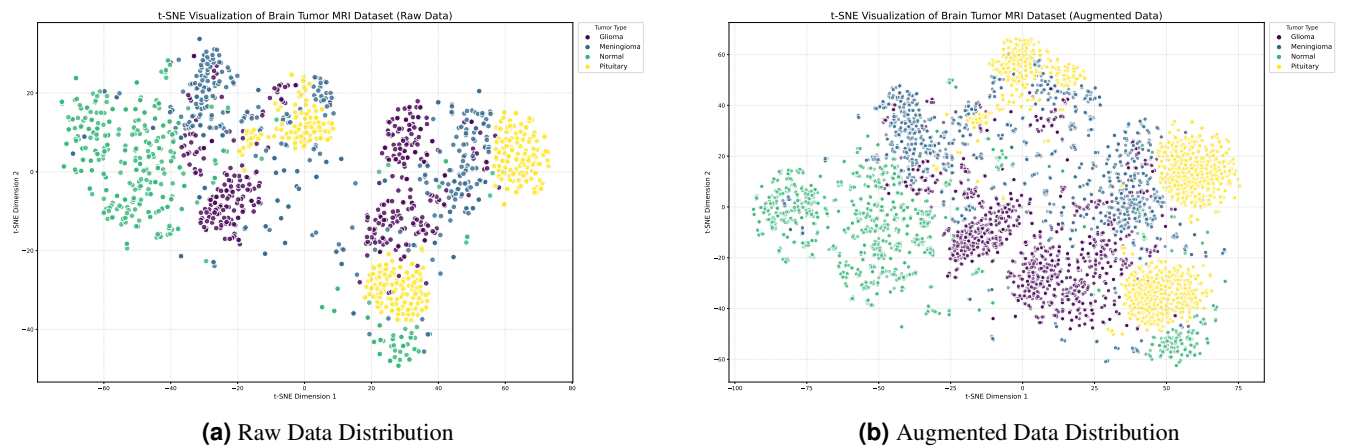


Figure 3. t-SNE visualization of the PMRAM dataset feature space. (a) The raw data distribution shows distinct but sparse clusters with inherent class overlaps, particularly between glioma and meningioma types. (b) The augmented data distribution demonstrates a significantly denser and more continuous manifold, filling gaps in the feature space to facilitate robust decision boundary learning.

Systematic architecture design

Teacher model selection

To secure a high-performance knowledge source, this study benchmarked 21 distinct deep learning architectures. Standard Convolutional Neural Networks (CNNs) like VGG16, ResNet, and DenseNet were part of the candidate models. They also featured modern, efficient architectures such as EfficientNet and MobileNet. Furthermore, Transformer-based models, including ViT and Swin Transformer, were evaluated. Based on empirical outcomes, EfficientNetB2 was identified as the teacher model f_T , presenting an optimal balance between diagnostic accuracy (99.67%) and parameter efficiency.

Student model derivation (CerebroNet)

The proposed student model, CerebroNet, was derived via systematic ablation rather than deploying an off-the-shelf lightweight network. The architecture is engineered to optimize gradient flow while minimizing parameter density. We engineered four “EnhancedConvBlock” modules in it, each featuring residual connections, Batch Normalization, and ReLU activation. A Channel Attention module²⁵ is embedded within each block to dynamically recalibrate feature channels. The network terminates with a global feature aggregation layer and a multi-layer perceptron classifier. This derivation yielded a compact architecture with 0.637 million parameters, specifically tailored for the distillation task. The architecture is illustrated in Fig. 4.

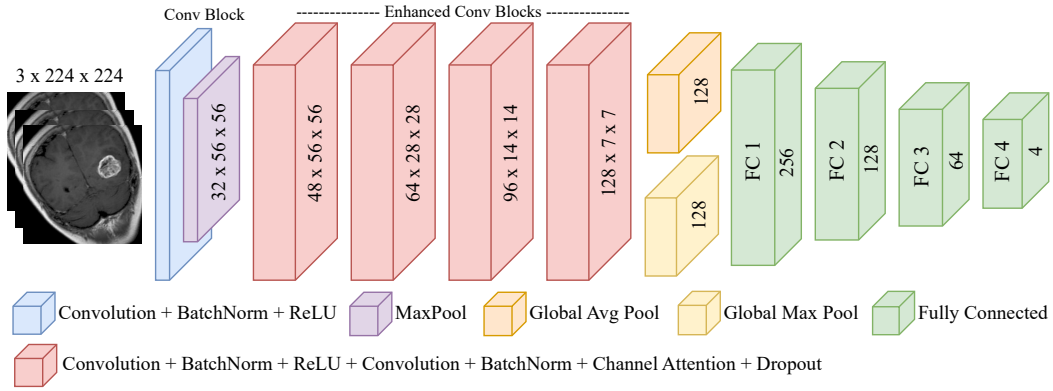


Figure 4. The 3D architecture of the proposed CerebroNet model. Derived through an ablation study, the model consists of an initial convolution followed by four EnhancedConvBlocks, each integrating a Channel Attention module for feature recalibration. The network concludes with global pooling and a fully connected classifier.

Framework to distill knowledge

We used knowledge distillation to pass on diagnostic proficiencies from the teacher model f_T to the student model f_S . The student is trained to minimize a composite loss function incorporating both ground-truth hard labels and soft probability distributions generated by the teacher. The distillation pipeline is visualized in Fig. 5.

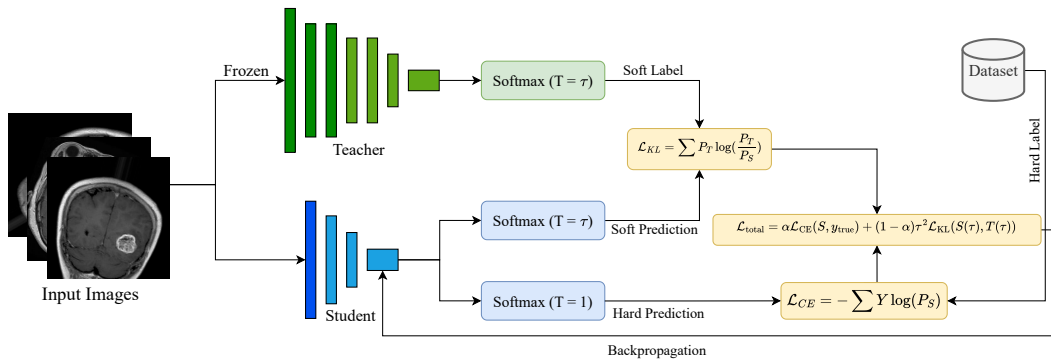


Figure 5. The knowledge distillation pipeline. The pre-trained teacher model (EfficientNetB2) processes an input image to produce soft labels. The lightweight student model (CerebroNet) also processes the same image. The total loss is a weighted sum of the standard cross-entropy loss (from hard labels) and the distillation loss (from soft labels).

The distillation loss is computed using the Kullback-Leibler (KL) divergence between the teacher and student logits, softened by a temperature parameter τ . The softened probability $p_i(\tau)$ for class i is defined as:

$$p_i(\tau) = \frac{\exp(z_i/\tau)}{\sum_j \exp(z_j/\tau)} \quad (1)$$

where z_i denotes the logits. The total loss function $\mathcal{L}_{\text{total}}$ aggregates the cross-entropy loss ($\mathcal{L}_{\text{CE}}$) and the KL divergence loss ($\mathcal{L}_{\text{KL}}$):

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{CE}}(S, y_{\text{true}}) + (1 - \alpha) \cdot \tau^2 \cdot \mathcal{L}_{\text{KL}}(S(\tau), T(\tau)) \quad (2)$$

Here, α acts as the weighting coefficient balancing the two loss components. The complete training procedure is formalized in Algorithm 1.

Algorithm 1 Knowledge Distillation Training for CerebroNet

Input: Teacher model f_T , Student model f_S , Training data D_{train} , Hyperparameters α, τ , Learning rate η

Output: Trained student model f_S

```

1: Initialize weights of  $f_S$  randomly
2: Freeze weights of  $f_T$ 
3: for each epoch do
4:   for each batch  $(X, y_{\text{true}})$  in  $D_{\text{train}}$  do
5:     Get teacher logits:  $T \leftarrow f_T(X)$ 
6:     Get student logits:  $S \leftarrow f_S(X)$ 
7:     Calculate cross-entropy loss:  $\mathcal{L}_{\text{CE}} \leftarrow \text{CrossEntropy}(S, y_{\text{true}})$ 
8:     Soften predictions:  $S(\tau) = \text{Softmax}(S/\tau)$ ,  $T(\tau) = \text{Softmax}(T/\tau)$ 
9:     Calculate distillation loss:  $\mathcal{L}_{\text{KL}} \leftarrow \text{KLDiv}(S(\tau), T(\tau))$ 
10:    Calculate total loss:  $\mathcal{L}_{\text{total}} \leftarrow \alpha \cdot \mathcal{L}_{\text{CE}} + (1 - \alpha) \cdot \tau^2 \cdot \mathcal{L}_{\text{KL}}$ 
11:    Update weights of  $f_S$  using gradient descent on  $\mathcal{L}_{\text{total}}$ 
12:   end for
13: end for
14: return  $f_S$ 

```

Experimental implementation and setup

This section outlines the experimental configuration, covering the implementation framework, optimization phases, and key hyperparameters employed to attain optimal performance.

Implementation framework

All models were implemented using the PyTorch framework and trained in a Kaggle environment utilizing NVIDIA T4 GPUs. To replicate the computational constraints characteristic of low-resource clinical hardware, training was strictly capped at 25 epochs. The models exhibited rapid convergence within this budget, confirming their applicability for deployment in environments with limited processing capacity.

Optimization phases

The optimization pipeline was executed in three distinct stages:

- **Teacher Benchmarking:** To establish a robust performance baseline, 21 teacher candidates were fine-tuned using the AdamW optimizer. A ReduceLROnPlateau scheduler was utilized, with early stopping patience set to 10 epochs to ensure convergence and prevent overfitting.
- **Student Optimization via Knowledge Distillation:** Following an architectural ablation study, the leading student candidates underwent rigorous hyperparameter tuning. The Optuna framework was employed to conduct independent 30-trial searches, specifically optimizing the distillation loss components (temperature τ and weighting coefficient α) alongside standard training parameters.
- **Pruning and Refinement:** To further decrease parameter density, 30% L1 unstructured global pruning was applied to the trained student model. This pruned architecture underwent a final fine-tuning phase of 10 epochs to recover any potential accuracy degradation. Inference time was measured on a single T4 GPU with a batch size of 1 to simulate real-world deployment latency.

Hyperparameters

Table 1 enumerates the optimized hyperparameters for the CerebroNet model, automatically determined via the Optuna framework. The learning rate was set to 0.002120, and the distillation temperature was optimized to 8.0 to facilitate effective soft target acquisition.

Table 1. Hyperparameters for CerebroNet training via knowledge distillation.

Hyperparameter	Value
Learning Rate	0.002120
Batch Size	32
Optimizer	AdamW
Weight Decay	1.00×10^{-4}
Weighting Coefficient (α)	0.38
Temperature (τ)	8.0
Training Epochs	25

Evaluation metrics

To evaluate the diagnostic performance of the applied models, standard classification metrics derived from the confusion matrix were utilized: Accuracy, Precision, Recall, and F1-score. For the detailed clinical evaluation on the raw dataset, Specificity and Negative Predictive Value (NPV) were additionally calculated. The formulas are:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (3)$$

$$\text{Precision or PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4)$$

$$\text{Recall or Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (5)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (7)$$

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} \quad (8)$$

Results

Analysis on augmented data

The primary objective of this study was to derive a model that optimizes the trade-off between computational efficiency and diagnostic accuracy. A comparative analysis of CerebroNet against 21 benchmarked architectures is presented in Table 2. The selected teacher model, EfficientNetB2, established a high performance ceiling with a test accuracy of 99.67%. In contrast, the derived student model, CerebroNet, achieved a test accuracy of 95.79% while utilizing only 0.637 million parameters. This represents a parameter reduction of approximately 91.7% relative to the teacher model (7.71 million parameters) and a significant deviation from other high-performing architectures such as ResNet50 (23.52 million) and ViT_B_16 (85.80 million).

Fig. 6 illustrates the relationship between parameter count and diagnostic accuracy on a logarithmic scale. CerebroNet occupies a distinct “high-efficiency zone” in the upper-left quadrant, demonstrating diagnostic capabilities comparable to substantially larger models while maintaining minimal architectural complexity. In terms of computational latency, CerebroNet recorded the lowest inference time of 0.437 ms per image on a T4 GPU, confirming its viability for real-time clinical deployment.

The training dynamics and optimization stability are depicted in Fig. 7. Fig. 7a shows the EfficientNetB2 teacher model exhibiting stable learning characteristics, reaching convergence without significant overfitting. Similarly, Fig. 7b demonstrates the student model’s robust convergence during the knowledge distillation process. The confusion matrices for both models, presented in Fig. 9, reveal class-specific diagnostic performance. The hyperparameter optimization process, conducted via Optuna, is visualized in Fig. 8, indicating that the learning rate and the distillation weighting coefficient (α) were the most influential factors governing model performance.

Table 2. Comprehensive performance comparison of all evaluated models on the independent test set, sorted by inference time.

Model	Parameters	Inference Time	Accuracy	Precision	Recall
CerebroNet	0.637M	0.437ms	95.79%	95.81%	95.79%
ShuffleNetV2_x1_0	1.26M	0.509ms	99.11%	99.07%	99.07%
SqueezeNet1_1	0.72M	0.571ms	63.64%	73.06%	64.21%
MobileNetV3_Large	4.21M	0.881ms	99.56%	99.53%	99.55%
MobileNetV2	2.23M	1.094ms	98.00%	97.97%	97.98%
EfficientNetB0	4.01M	1.433ms	99.00%	98.95%	98.99%
ResNet34	21.29M	1.697ms	93.79%	93.95%	93.58%
EfficientNetB1	6.52M	1.930ms	97.45%	97.42%	97.33%
EfficientNetB2	7.71M	2.114ms	99.67%	99.65%	99.65%
InceptionV3	24.35M	2.155ms	99.00%	98.97%	98.98%
ResNet50	23.52M	2.954ms	88.36%	89.16%	87.86%
DeiT_Small	21.67M	2.988ms	93.79%	93.67%	93.69%
DenseNet121	6.96M	3.110ms	92.24%	92.64%	92.52%
ConvNeXt_Tiny	27.82M	3.812ms	59.65%	58.69%	58.95%
ResNet101	42.51M	5.439ms	92.13%	92.13%	92.01%
VGG16	134.28M	5.632ms	58.54%	58.81%	58.65%
MaxViT_Tiny	30.41M	7.262ms	99.45%	99.42%	99.41%
ConvNeXt_Small	49.46M	7.475ms	67.29%	67.69%	66.85%
DenseNet161	26.48M	7.535ms	98.45%	98.40%	98.44%
SwinT_Small	48.84M	7.984ms	96.67%	96.64%	96.63%
ViT_B_16	85.80M	11.903ms	73.73%	75.29%	73.70%
SwinT_Base	86.75M	12.918ms	98.34%	98.33%	98.33%

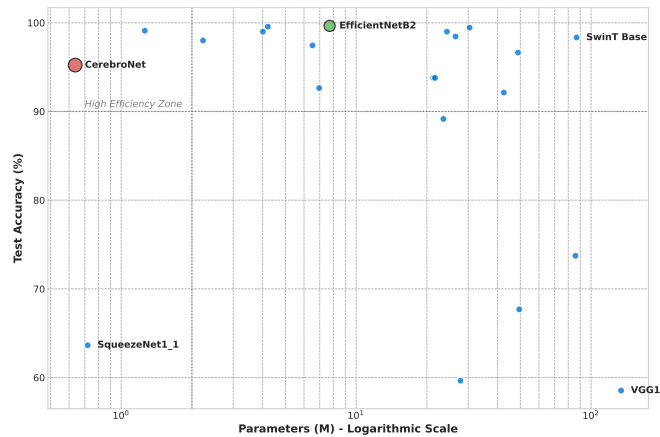
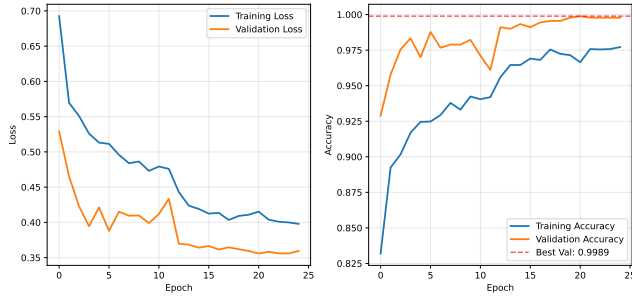


Figure 6. Model performance vs. parameter count. Test accuracy of all 21 benchmarked models is plotted against their parameter count (logarithmic scale). The plot highlights CerebroNet’s unique position in the top-left quadrant, demonstrating its ability to achieve high diagnostic accuracy while maintaining an exceptionally low parameter count compared to other high-performing models.

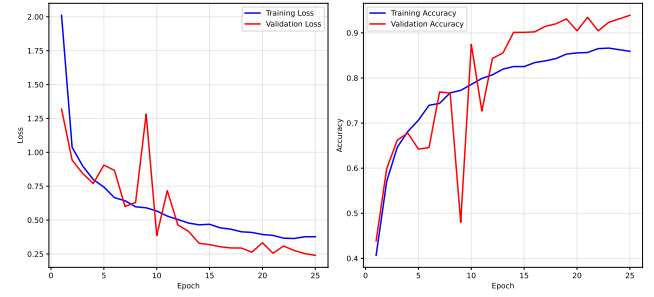
Analysis on raw data

To assess the clinical applicability of CerebroNet beyond the augmented training distribution, an evaluation was performed on the complete raw PMRAM dataset (1,505 images). The model attained an overall accuracy of 96.21%. Table 3 details the one-vs-rest clinical performance metrics. The model showed a specificity of 98.73% and a Negative Predictive Value (NPV) of 98.74%. These metrics are important for reducing false positives in a clinical setting.

Statistical stability was verified through a bootstrap analysis comprising 1,000 iterations. As shown in Table 4, the 95% confidence intervals for class-wise precision, recall, and F1-score confirm consistent performance. The overall accuracy was bounded within the interval [95.22% – 97.14%]. Furthermore, Table 5 presents a comparison with the current state-of-the-art

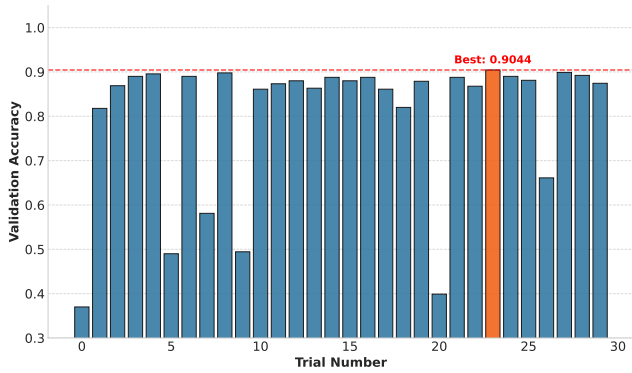


(a) Teacher model (EfficientNetB2)

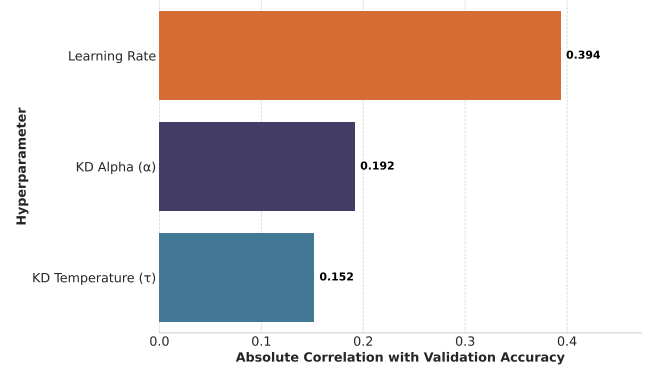


(b) Student model (CerebroNet)

Figure 7. Training and validation accuracy and loss curves. (a) The teacher model (EfficientNetB2) exhibits stable learning without significant overfitting, reaching a high validation accuracy. (b) The proposed student model (CerebroNet) demonstrates stable convergence throughout the knowledge distillation process.

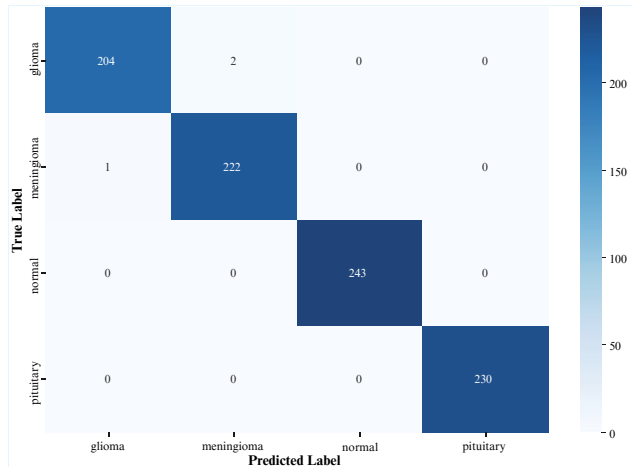


(a) Optimization History

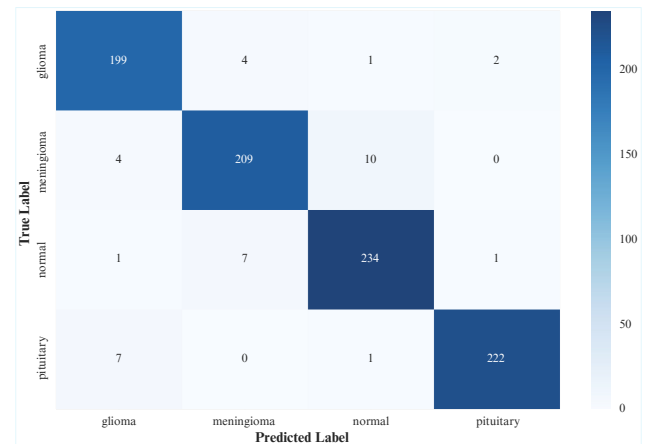


(b) Hyperparameter Importances

Figure 8. Optuna hyperparameter search process for CerebroNet. (a) The validation accuracy was achieved across 30 trials, with the best-performing trial highlighted. (b) The relative importance of each hyperparameter indicates that the learning rate and knowledge distillation alpha (α) were the most influential factors in model performance.



(a) Teacher model (EfficientNetB2)



(b) Student model (CerebroNet)

Figure 9. Confusion matrices for (a) EfficientNetB2 and (b) CerebroNet (Proposed), evaluated on the test set.

on the PMRAM dataset. CerebroNet outperformed the optimized ResNet-18 model proposed by He et al.²¹, achieving a higher accuracy (96.21% vs. 95.69%) while reducing parameter count by a factor of 17.

Table 3. Detailed one-vs-rest clinical performance metrics of CerebroNet on the raw PMRAM dataset (1505 images).

Class	Sensitivity (Recall)	Specificity	PPV (Precision)	NPV	F1-Score
Glioma	0.9491	0.9885	0.9646	0.9833	0.9568
Meningioma	0.9421	0.9877	0.9607	0.9817	0.9513
Normal	0.9773	0.9811	0.9485	0.9918	0.9627
Pituitary	0.9786	0.9920	0.9759	0.9929	0.9772

Table 4. Performance metrics with 95% confidence intervals for CerebroNet on the raw PMRAM dataset, determined via bootstrapping (1000 iterations).

Class	Precision	Recall	F1-Score	Support
Glioma	0.94-0.98	0.93-0.97	0.94-0.97	373
Meningioma	0.94-0.98	0.92-0.96	0.93-0.97	363
Normal	0.93-0.97	0.96-0.99	0.95-0.98	396
Pituitary	0.96-0.99	0.96-0.99	0.96-0.99	373
Accuracy	96.21% [95.22% – 97.14%]			1505

Table 5. Comparison with state-of-the-art on the PMRAM dataset.

Model	Source	Parameters (M)	Accuracy (%)
ResNet-18 (GSECA)	He et al. ²¹	~11	95.69
CerebroNet	This Study (Proposed)	0.637	96.21

Analysis on external data

To evaluate the model’s robustness across diverse data distributions, validation was conducted on the independent “Brain Tumor Data” dataset³⁷. CerebroNet yielded an overall accuracy of 91.00% after minimal fine-tuning. The classification report in Table 6 indicates balanced performance across all diagnostic classes, with high precision and recall values, notably 99% recall for the “No Tumor” class. This validates the effective transfer of learned feature representations to external clinical environments.

Table 6. Classification report for CerebroNet on the “Brain Tumor Data” dataset.

Class	Precision	Recall	F1-Score	Support
Glioma	0.95	0.90	0.93	300
Meningioma	0.86	0.75	0.80	306
No Tumor	0.90	0.99	0.94	405
Pituitary	0.90	0.96	0.93	300
Accuracy			91.00%	1311
Macro Avg	0.91	0.90	0.90	1311
Weighted Avg	0.90	0.91	0.90	1311

Explainable AI analysis

Layer-CAM visualization was employed to compare the decision-making logic of the teacher and student models. Fig. 10 presents a qualitative comparison across four scenarios. In the “Meningioma” case (Fig. 10c), both models exhibit nearly identical attention maps, indicating successful knowledge transfer. Conversely, discrepancies are observed in the “Pituitary” case (Fig. 10d), where the student model seemingly displays more diffuse activation compared to the highly localized focus of the teacher. This analysis provides transparency regarding the trade-offs in visual reasoning incurred during model compression.

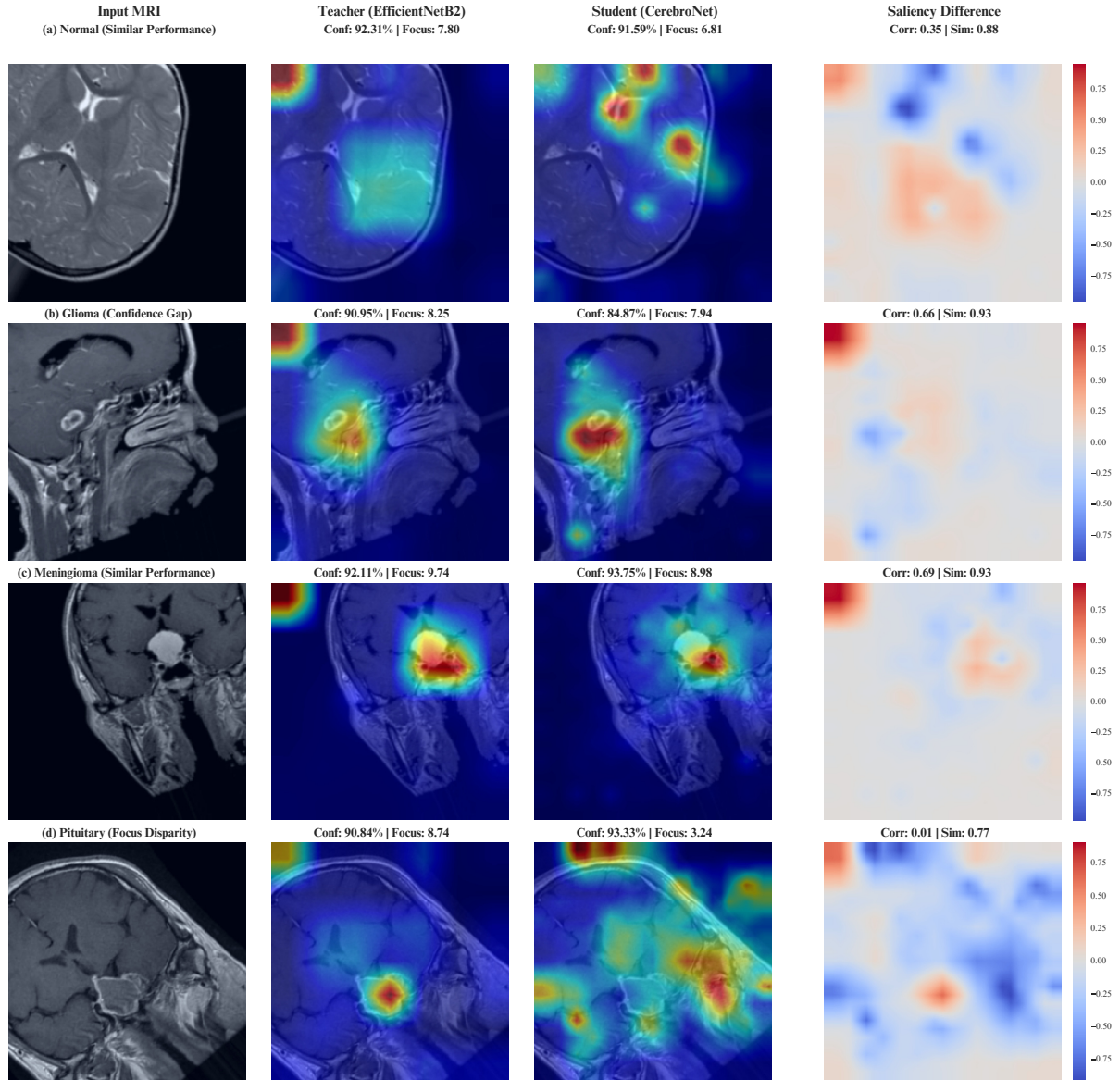


Figure 10. Nuanced Layer-CAM analysis of the teacher (EfficientNetB2) and student (CerebroNet) models across all four diagnostic classes. **(a) Normal:** A case of similar performance where both models correctly show diffuse, low-intensity activations, indicating no specific region of pathology. **(b) Glioma:** An example of a confidence gap, where both models correctly identify the tumor, but the teacher does so with higher confidence (90.95%) compared to the student (84.87%). **(c) Meningioma:** A case demonstrating highly successful knowledge transfer, where both models’ attention maps and high-confidence estimations are nearly identical. **(d) Pituitary:** A case highlighting a focus disparity, where the student model’s attention is significantly more diffuse and less localized on the tumor compared to the teacher’s highly focused activation. This detailed comparison validates the general success of the knowledge transfer while honestly illustrating the subtle performance trade-offs inherent in model compression.

Ablation study

We performed a series of ablation experiments to quantify the impact of architectural components and optimization stages. Fig. 11a illustrates the component contribution analysis, revealing that the integration of Channel Attention alone improved validation accuracy from 56.78% to 83.44%. In contrast, Spatial Attention yielded lower gains. Subsequently, a head-to-head comparison via knowledge distillation (Fig. 11b) confirmed that the Channel Attention variant outperformed the combined attention model (95.23% vs. 93.02%). Finally, the impact of unstructured pruning is depicted in Fig. 11c, where the application of 30% L1 pruning increased test accuracy to 95.79%, establishing the pruned Channel Attention variant as the optimal

configuration for CerebroNet.

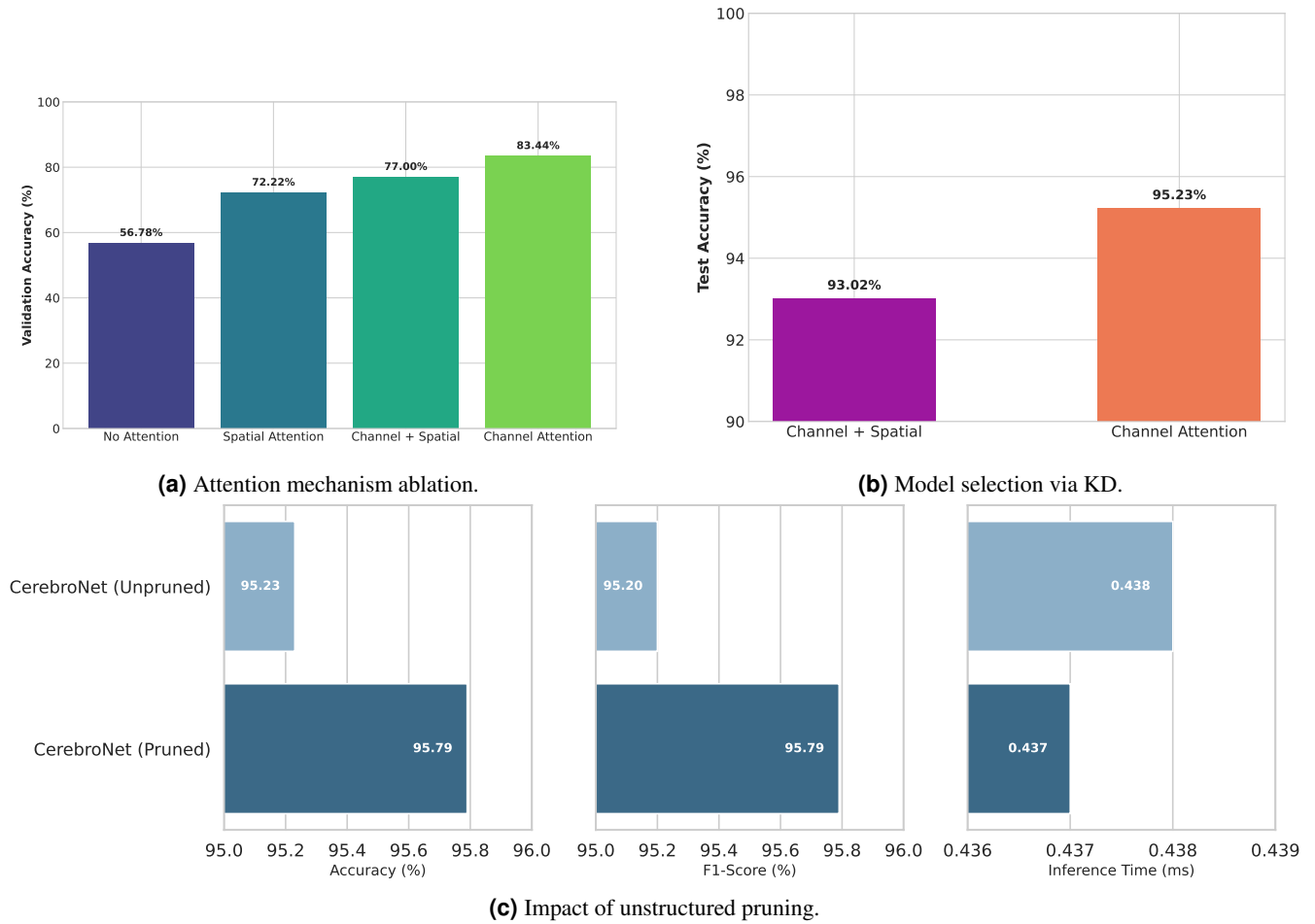


Figure 11. The complete ablation study for CerebroNet refinement. (a) The initial architecture study revealed channel attention as the most effective mechanism. (b) A head-to-head comparison using knowledge distillation confirmed the channel-attention-only variant as superior. (c) The final model was further optimized via unstructured pruning, which improved accuracy and F1-score while reducing inference time.

Discussion

The results from this study seem to strongly support the effectiveness of the proposed optimization framework. CerebroNet successfully assimilated complex diagnostic representations from the teacher architecture, achieving a test accuracy of 95.79% on augmented data and increasing to 96.21% on the raw, un-augmented clinical dataset. This consistency across datasets is important. It suggests that the model learned real pathological features instead of just fitting to specific artifacts from augmentation. A significant finding from the ablation study challenges the prevailing assumption that complex attention mechanisms inherently yield superior performance; empirical results demonstrated that a streamlined Channel Attention module provided the optimal balance of computational cost and feature recalibration for this specific lightweight architecture. Furthermore, the validation of the model on the PMRAM dataset¹⁴ addresses a critical gap in demographic representation, ensuring diagnostic equity for South Asian populations often underrepresented in Western-centric AI research. When compared to the optimized ResNet-18 model proposed by He et al.²¹ on the same cohort, CerebroNet not only achieved superior diagnostic accuracy (96.21% versus 95.69%) but did so with a 17-fold reduction in parameter density (0.637 million versus ~11 million). This extreme efficiency, characterized by a memory footprint under 5 MB and an inference latency of 0.437 ms, underscores the model's suitability for integration into standard clinical PACS workflows or edge devices without necessitating high-performance GPU infrastructure³⁸.

While the proposed framework demonstrates significant potential for resource-constrained diagnostics, several limitations

must be acknowledged. First, the explicit prioritization of single-channel MRI to maximize deployment feasibility precludes the utilization of multi-modal diagnostic signals (e.g., T1, T2, and FLAIR fusion), which may restrict diagnostic sensitivity in complex cases where cross-modality correlation is decisive³⁹. Second, the aggressive parameter reduction strategy introduces an inherent capacity trade-off; while the model excels at core diagnostic tasks, its ability to generalize to extremely rare or “long-tail” pathological presentations compared to over-parameterized networks requires further investigation. Finally, as this study relies on retrospective validation, the findings function as a proof-of-concept, and true clinical utility must ultimately be verified through prospective clinical trials in diverse healthcare settings.

Conclusion

Classifying brain tumors is still a major global health issue, particularly in areas with limited resources. Here, the challenges of diagnosis are increased by poor infrastructure and biases in the training data. This study tackles these problems by creating a clear optimization framework to develop “CerebroNet,” a lightweight and transparent deep learning classifier. Leveraging the PMRAM dataset, which offers a diverse representation of South Asian demographics, we established a rigorous pipeline that refined diagnostic capability from a high-performance EfficientNetB2 teacher into a compact student architecture through systematic ablation, knowledge distillation, and unstructured pruning. This approach achieved a significant reduction in computational requirements, with CerebroNet utilizing only 0.637 million parameters, a 17-fold reduction compared to state-of-the-art benchmarks, while maintaining superior diagnostic fidelity. The model demonstrated robust performance, achieving 96.21% accuracy on raw clinical data and 91% on an external dataset, thereby validating its generalization capabilities across diverse imaging conditions. Furthermore, the integration of Explainable AI (XAI) techniques provided critical transparency, confirming that the student model retains the salient visual reasoning of its teacher. Collectively, these findings demonstrate that systematic model compression can effectively harmonize computational efficiency with diagnostic precision, offering a scalable pathway for deploying reliable AI tools in healthcare environments with limited resources.

Future directions involve extending this framework to multi-institutional and multi-ethnic cohorts to further validate the model’s generalizability on a global scale. We believe integrating multi-modal MRI sequences, such as T1, T2, and FLAIR, could enhance diagnostic sensitivity for complex pathological presentations without compromising the model’s efficiency. And the most important point is, we need future clinical trials to measure the real-world effects of CerebroNet on diagnostic processes with physician validation⁴⁰.

Data availability

The datasets analyzed during the study are available in the Mendeley Data repository. They can be accessed through <https://doi.org/10.17632/m7w55sw88b.1> and <https://data.mendeley.com/datasets/w4sw3s9f59/1>.

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Author contributions

All authors conceived the experiment(s). U.H. conducted the experiment(s). All authors analyzed the results. U.H. wrote the main manuscript text and prepared all the figures. M.A.N. and R.K. reviewed and edited the manuscript. All authors reviewed the final manuscript.

Competing interests

The authors declare no competing interests.